FACULTY OF ENGINEERING AND TECHNOLOGY

Course Code: CEF 440

Group 18

Design and Implementation of an Adaptive E-learning Platform based on Quality of Experience for Inclusive Online Education in Low-Resource Settings

Course Title: Internet Programming (J2EE) and Mobile Programming

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Introduction

System modelling and design is an important phase in software engineering because it provides a structured representation of how a system operates, how users interact with it, and how different software and hardware components communicate. For this project, the system modelling process focuses on the design of an Adaptive E-Learning Platform developed for inclusive online education in low-resource environments.

The platform is designed to support students and instructors in regions with unstable internet connectivity and limited digital resources. The system uses adaptive streaming, offline storage, synchronization mechanisms, and Quality of Experience (QoE) monitoring to ensure that learning remains accessible even under poor network conditions.

This report presents the major system modelling diagrams used to describe the structure, behavior, data movement, and deployment architecture of the platform. The diagrams included in this report are:

- Context Diagram
- Data Flow Diagram (DFD)
- Use Case Diagram
- Sequence Diagram
- Class Diagram
- Deployment Diagram

Each section explains the purpose of the diagram and how it contributes to the overall system design.

1. Context Diagram

1.1 Overview

A Context Diagram is a high-level system representation used in software engineering to define the boundaries of a system and identify the external entities that interact with it. It focuses on the overall communication between the system and external actors without exposing internal processes or database structures.

The context diagram for the Adaptive E-Learning Platform represents the application as a single process that communicates with external users and environments. The platform is specifically designed for low-resource environments where internet access is unstable and mobile devices may have limited performance capabilities.

The primary objective of the system is to deliver educational content efficiently by monitoring network conditions and dynamically adjusting media quality to reduce buffering and improve accessibility.

1.2 External Entities and Data Flows

A. Instructor

The instructor is responsible for managing academic content within the platform.

Inputs to the System

- Authentication credentials
- Course materials
- Lecture schedules
- Quizzes and assignments

Outputs from the System

- Login confirmation
- Upload confirmation
- Course management feedback

B. Network Environment

The network environment represents the internet infrastructure monitored by the application.

Inputs to the System

- Bandwidth speed
- Latency metrics
- Packet loss statistics
- Connectivity quality indicators

Outputs from the System

The platform does not directly send information back to the network environment. Instead, it continuously monitors network conditions to optimize content delivery.

C. Student

The student is the primary consumer of the learning platform.

Inputs to the System

- Login credentials
- Course requests
- Interaction logs
- Assignment submissions
- Quiz responses

Outputs from the System

- Adaptive course content
- Content recommendations
- Notifications and updates
- Quiz results and feedback

The system dynamically modifies media delivery depending on the detected network conditions. For example, when bandwidth becomes poor, the application may switch from high-quality video streaming to audio-only or text-based learning materials.

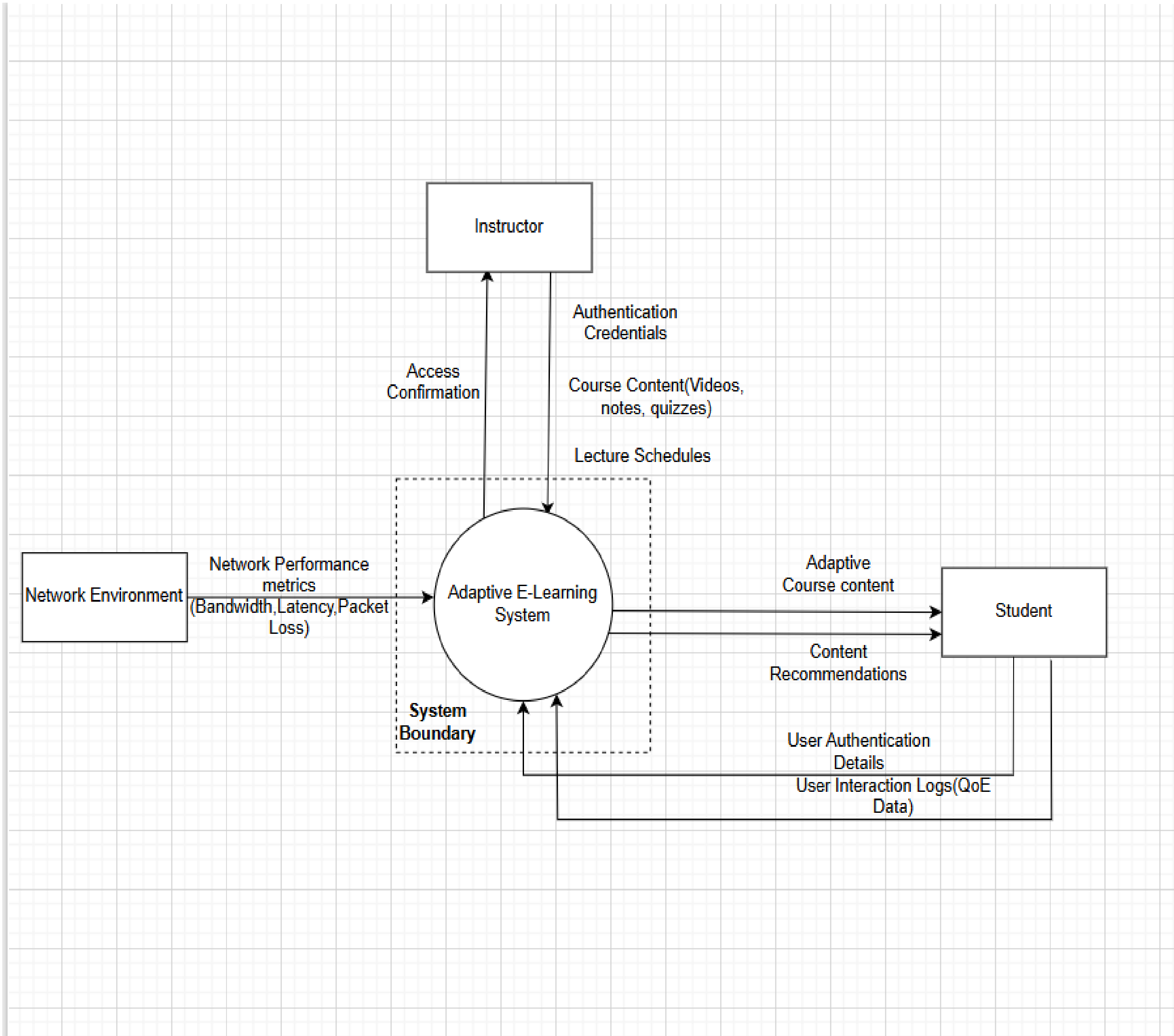

Figure 1: Context Diagram

The model below demonstrates how the platform collects learning materials from instructors, monitors network conditions, and delivers optimized educational content to students in low-connectivity environments.

2. Data Flow Diagram (DFD)

2.1 Overview

The Data Flow Diagram (DFD) illustrates how data moves within the Adaptive E-Learning Platform. Unlike the context diagram, which provides a high-level external view, the DFD explains the internal processing logic and how information flows between processes, users, and storage components.

The system is designed to support learning continuity in low-bandwidth environments by monitoring internet quality, managing course content adaptively, and synchronizing offline learning progress.

2.2 Major Internal Processes

Process 1.0 – Authenticate User

This process handles user authentication and authorization.

Functions

- Receives user login credentials
- Verifies user identity
- Communicates with the Moodle API
- Stores authentication tokens securely

Process 2.0 – Monitor Network Speed

This process continuously evaluates the quality of the internet connection.

Functions

- Collects connectivity information
- Measures bandwidth and latency
- Detects network availability
- Produces network quality status information

Process 3.0 – Manage Course Content

This is the core adaptive processing component of the platform.

Functions

- Retrieves course materials from Moodle
- Receives uploaded files from instructors
- Adapts media quality based on network conditions
- Streams optimized educational content
- Stores student learning progress locally

2.3 Data Storage

The platform uses local NoSQL storage mechanisms to support offline learning.

Hive Local Storage

This data store is responsible for:

- Saving offline course content
- Caching user progress
- Preserving quiz attempts
- Synchronizing data when internet connectivity is restored

2.4 Data Dictionary Summary

Data Flow	Description
Raw Connection Stream	Real-time connectivity metrics received from the mobile device
Network Quality Status	Processed network condition classification
Local Cached Progress	Offline learning progress stored locally
Course Materials Upload	Learning materials uploaded by instructors
Raw Course Payload	Course information retrieved from Moodle
Optimized Media Stream	Adaptively streamed educational content

2.5 Validation of the DFD

The DFD structure was validated to ensure compliance with systems analysis principles.

Validation Checks

- Balanced data inputs and outputs
- Proper process transformation logic
- Restricted direct access to data stores
- Consistent naming conventions across the system
- Proper separation between external entities and internal processes

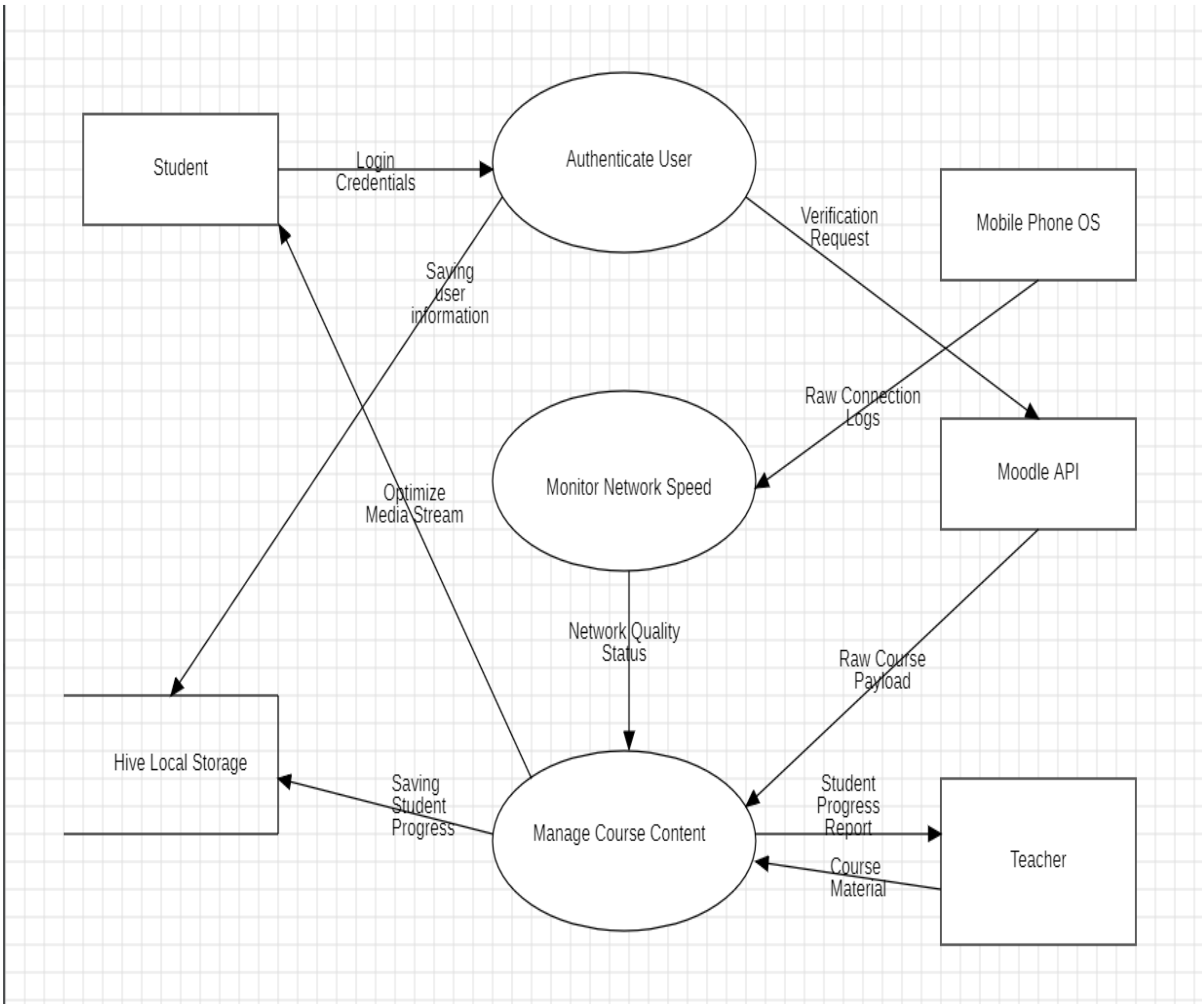

Figure 2: Level 1 Data Flow Diagram

The Data Flow Diagram demonstrates how the platform processes information internally to provide adaptive learning experiences. Through authentication, network monitoring, and intelligent content management, the system ensures efficient educational delivery even under poor connectivity conditions.

3. Use Case Diagram

3.1 Overview

The Use Case Diagram models the interactions between users and the Adaptive E-Learning Platform. It identifies the major system actors and describes the functional services available to each user category.

The diagram serves as a behavioral model that connects the system requirements to the activities performed by students, instructors, administrators, and automated background services.

3.2 System Actors

Student

The student is the primary user of the system.

Main Activities

- Register and log in
- Browse assigned courses
- Download content for offline learning
- Submit assignments
- Take quizzes
- Receive notifications
- View learning progress
- Enable data saver mode

Instructor

The instructor manages learning resources and student engagement.

Main Activities

- Upload learning materials
- Create quizzes and assignments
- Monitor student progress
- Communicate with students
- Provide academic feedback

Administrator

The administrator oversees system operations.

Main Activities

- Manage user accounts
- Monitor platform activity
- Configure system settings
- Supervise technical operations

Automated Monitoring System

The automated monitoring system continuously performs adaptive operations.

Main Activities

- Monitor connection quality
- Adjust media streaming quality
- Synchronize offline data
- Perform background downloads
- Manage local cache states

3.3 Functional Modules

Authentication and Security

- User registration
- Login and logout
- Password recovery
- Credential verification

Learning and Content Access

- Browse courses
- Download materials
- Stream adaptive media
- Access quizzes and assignments

Communication and Feedback

- Push notifications

- Teacher-student messaging
- Performance tracking
- Learning analytics

Adaptive Network Management

- Connection monitoring
- Adaptive bitrate streaming
- Offline synchronization
- Background optimization tasks

3.4 Key Relationships

The use case model includes dependency relationships between critical system functions.

Include Relationships

- Stream Content includes Monitor Connection and Speed
- Synchronize Progress includes Monitor Connection and Speed

These dependencies ensure that adaptive operations always rely on current network conditions before executing streaming or synchronization tasks.

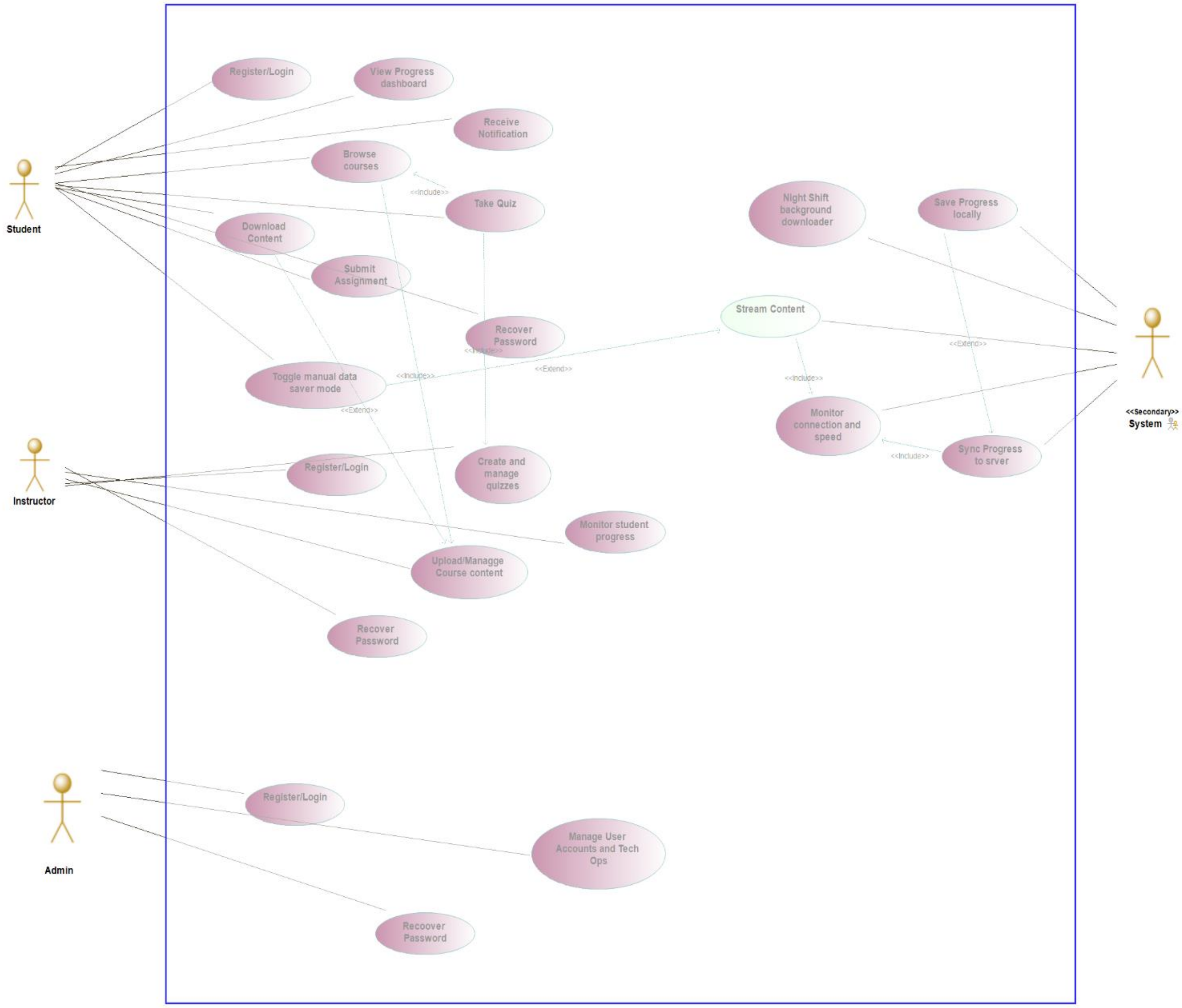

Figure 3: Use Case Diagram

The Use Case Diagram provides a clear representation of the services offered by the Adaptive E-Learning Platform and the interactions between system actors and functional modules. The model emphasizes adaptive learning delivery, offline accessibility, and efficient communication between users and the system.

4. Sequence Diagram

4.1 Introduction

The Sequence Diagram illustrates the chronological interaction between users, automated services, and the core system during execution. It demonstrates how the Adaptive E-Learning Platform coordinates authentication, content delivery, synchronization, and adaptive streaming processes.

The diagram highlights how the platform maintains accessibility and learning continuity in environments with unstable internet connectivity.

4.2 Main Components

Component	Role
Student	Accesses learning content and interacts with the platform
Instructor	Uploads materials and evaluates student submissions
Administrator	Manages technical operations and user profiles
AI Monitoring System	Monitors bandwidth and adjusts media quality
System Engine	Coordinates synchronization and optimization tasks
Core System	Handles authentication, storage, and communication

4.3 Sequence Flow Description

Step 1 – Authentication

The student registers or logs into the system. The system validates the credentials and initializes a secure learning session.

Step 2 – Content Access

The instructor uploads course materials, and the student browses assigned learning resources through the platform.

Step 3 – Offline Learning Support

Students download learning materials for offline access. The platform stores content locally for future use.

Step 4 – Assessment and Feedback

Students submit assignments and complete quizzes. The instructor evaluates submissions and provides feedback through the system.

Step 5 – Adaptive Streaming

The AI monitoring module evaluates network conditions and dynamically adjusts media quality. Depending on bandwidth conditions, the platform may switch between high-quality video, low-resolution video, audio-only, or text-based delivery.

Step 6 – Synchronization

Learning progress is stored locally when internet connectivity is unavailable. Once connectivity improves, synchronization processes automatically update the server.

Step 7 – Administrative Operations

Administrators manage user profiles, monitor technical activities, and supervise platform performance.

Step 8 – Background Optimization

The platform performs background downloads during off-peak network periods to reduce bandwidth costs and improve learning accessibility.

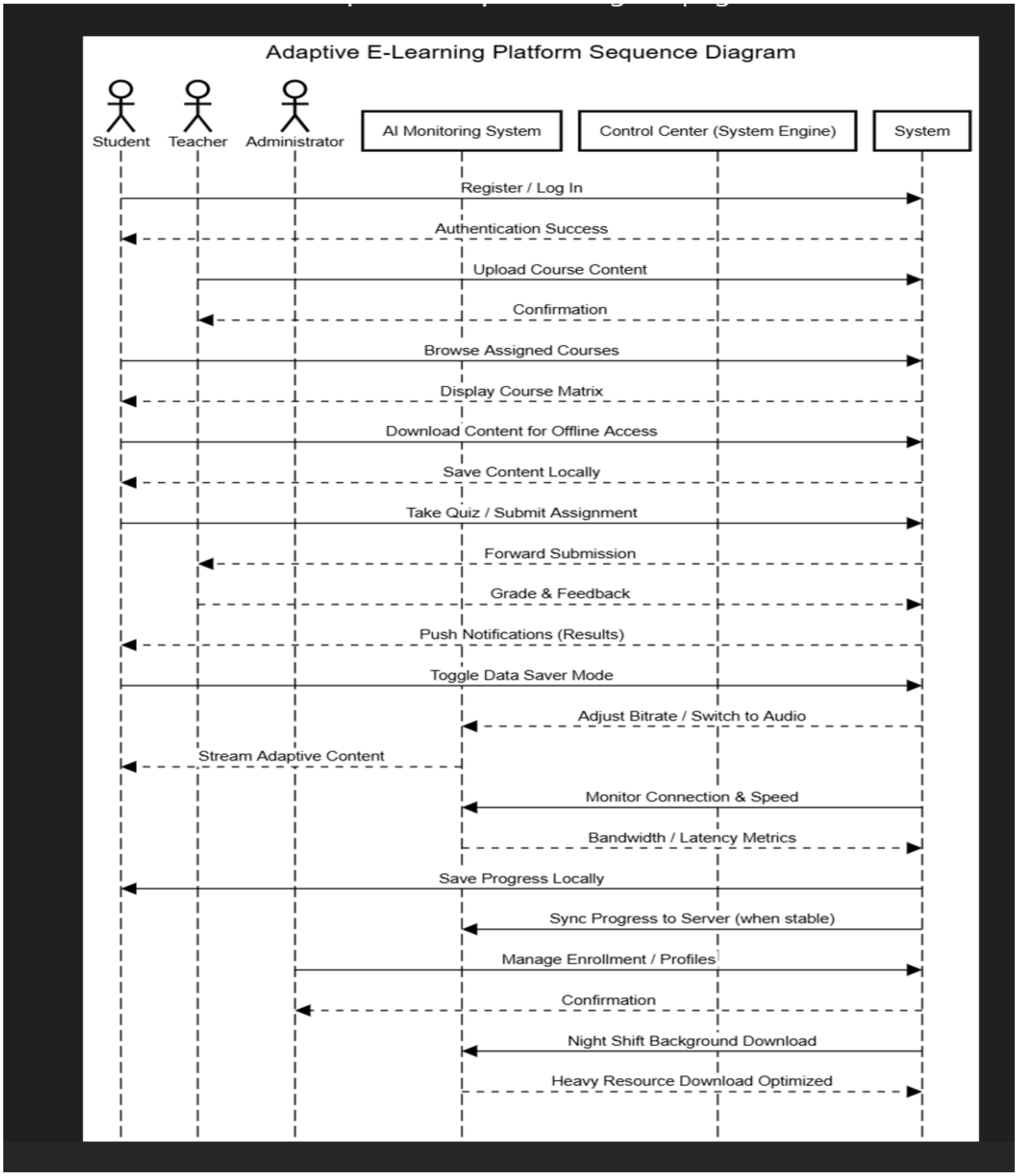

Figure 4: Sequence Diagram

The Sequence Diagram demonstrates the dynamic interactions that occur during system operation. It highlights the integration of adaptive streaming, offline support, synchronization, and intelligent monitoring mechanisms that ensure continuous learning experiences for students in low-resource environments.

5. Class Diagram

5.1 Overview

The Class Diagram describes the object-oriented structure of the Adaptive E-Learning Platform. It defines the system classes, their attributes, methods, and relationships.

The model provides a blueprint for software implementation by showing how the major components of the platform interact.

5.2 User Hierarchy

User Class

The User class acts as the parent class for all system users.

Common Attributes

- User ID
- Name
- Email
- Password

Common Methods

- Login()
- Logout()

Student Class

The Student class extends the User class.

Main Functions

- Enroll in courses
- Download content
- Submit assignments
- Take quizzes
- View progress

Instructor Class

The Instructor class extends the User class.

Main Functions

- Upload course materials
- Create assessments
- Monitor student progress
- Send notifications

Administrator Class

The Administrator class manages system-level activities.

Main Functions

- Manage users
- Configure settings
- Monitor platform performance
- Access analytics

5.3 Core Learning Components

Course Class

The Course class stores course-related information.

Responsibilities

- Manage course modules
- Store learning materials
- Track enrolled students

Quiz Class

The Quiz class manages assessments.

Responsibilities

- Store questions and answers
- Manage quiz attempts
- Record student scores

5.4 Supporting Technical Modules

Connectivity Module

This module monitors internet connection quality and network speed.

Media Player

This component handles adaptive media playback and data-saving operations.

Offline Storage

This module manages local data caching, synchronization, and encryption.

Notification Module

This component manages alerts, reminders, and queued notifications.

5.5 Relationships Between Classes

Inheritance

Inheritance relationships allow specialized user classes to extend the general User class.

Association

Association relationships show how users interact with courses, quizzes, and notifications.

Dependency

Dependency relationships represent temporary interactions between system modules.

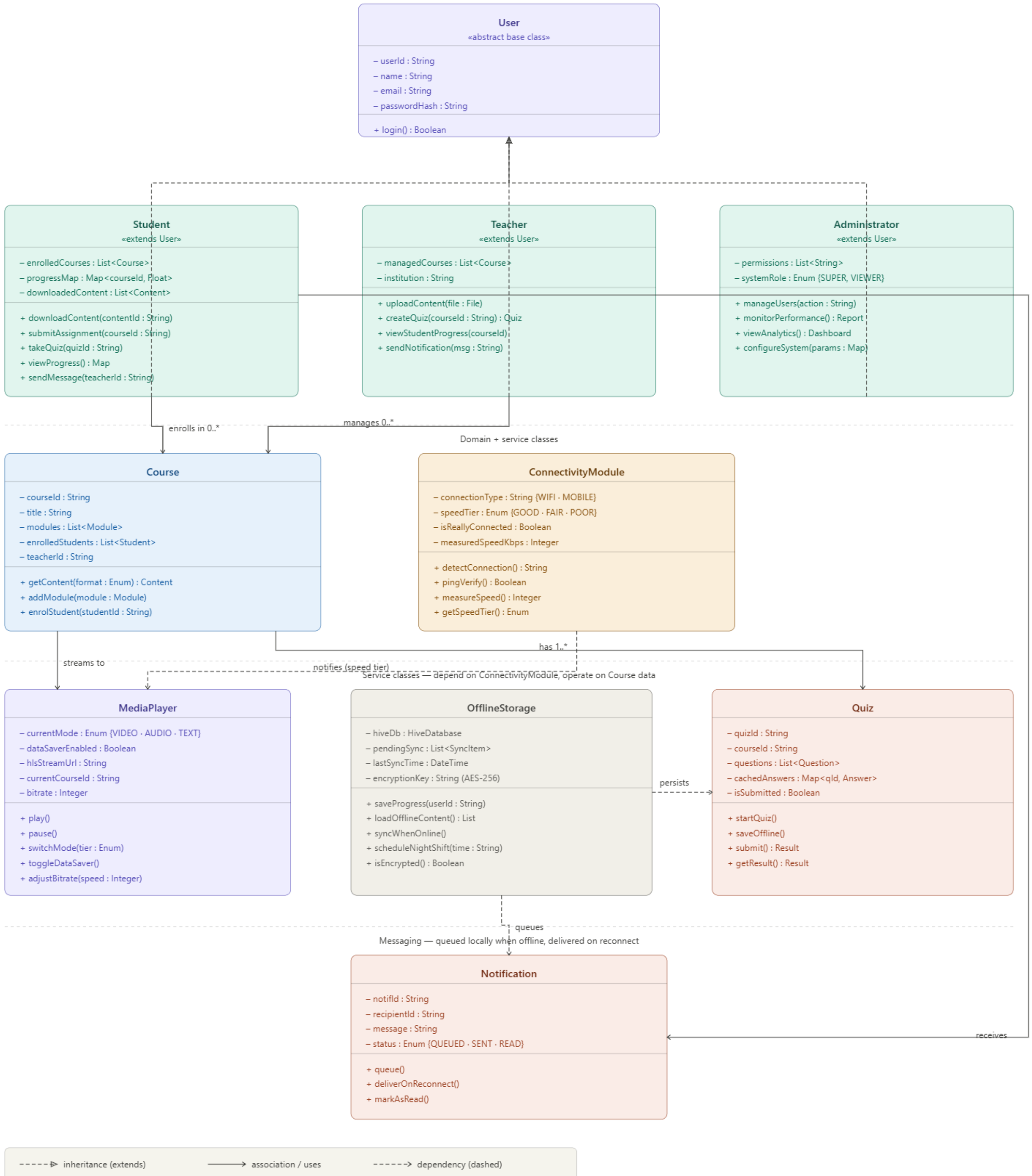

Figure 5: Class Diagram

The Class Diagram provides a detailed representation of the internal software architecture of the Adaptive E-Learning Platform. The design promotes modularity, scalability, maintainability, and efficient management of learning services.

6. Deployment Diagram

6.1 Overview

The Deployment Diagram illustrates the physical architecture of the Adaptive E-Learning Platform. It describes how software components are distributed across hardware devices, cloud infrastructure, and external services.

The deployment architecture is designed to provide reliable educational access in low-connectivity environments while supporting secure communication, adaptive streaming, and offline synchronization.

6.2 Major Deployment Components

Android Mobile Application

The mobile application is used primarily by students and instructors.

Features

- Built using Flutter
- Supports offline learning
- Uses encrypted Hive local storage
- Synchronizes data when internet access is restored

Web Browser Interface

The browser interface is used by instructors and administrators.

Features

- Course management
- Dashboard monitoring
- Administrative operations
- Secure API communication

Cloud Application Server

The cloud server acts as the main processing environment.

Features

- Built using Node.js
- Handles authentication and synchronization
- Processes adaptive media streaming
- Converts videos into multiple resolutions

Moodle LMS Server

The Moodle server stores academic data.

Features

- User management
- Course storage
- Quiz and grade management
- Learning analytics

Cloudflare CDN

The Content Delivery Network improves media accessibility.

Features

- Faster media delivery
- Reduced buffering
- Cached streaming resources
- Improved performance in Sub-Saharan Africa

6.3 Communication Technologies

The system components communicate using:

- HTTPS
- REST APIs
- HLS adaptive streaming

6.4 System Workflow

- Mobile and web clients send requests to the cloud application server.
- The cloud server retrieves academic data from Moodle.
- Media files are streamed through the Cloudflare CDN.

- Offline data is synchronized automatically when connectivity becomes available.

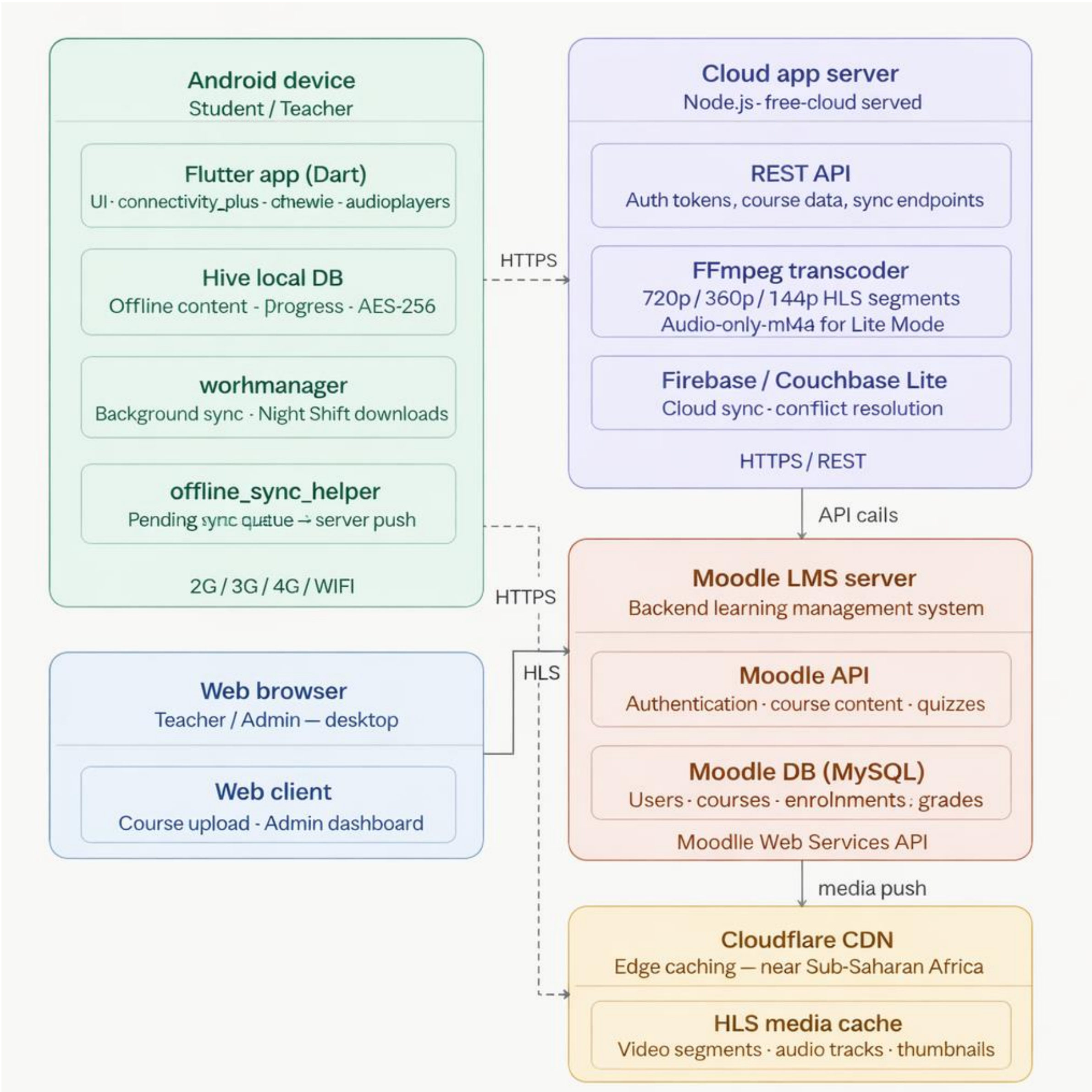

Figure 6: Deployment Diagram

The Deployment Diagram demonstrates how the Adaptive E-Learning Platform integrates mobile applications, cloud infrastructure, learning management systems, and content delivery services to provide secure, scalable, and reliable online learning experiences.

Conclusion

The system modelling and design process provides a comprehensive representation of the Adaptive E-Learning Platform and its operational architecture. Through the use of Context, Data Flow, Use Case, Sequence, Class, and Deployment diagrams, the report successfully captures the structural, behavioral, and deployment aspects of the system.

The proposed platform addresses major educational challenges in low-resource environments by integrating adaptive streaming, offline synchronization, network-aware optimization, and intelligent content delivery mechanisms. The system architecture emphasizes scalability, accessibility, maintainability, and user-centered learning experiences.

Overall, the diagrams and accompanying explanations provide a solid blueprint for the implementation and future expansion of the Adaptive E-Learning Platform.